# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** $\frac{3}{4}$ | Substitute the standard values |
| 2 | **(c)** 3 | $\frac{c}{q}+\frac{p}{q}$ |
| 3 | **(c)** \frac{1}{2} | $\frac{c}{p}-\frac{q}{p}$ |
| 4 | **(b)** 2 | Each ratio is 1 |
| 5 | **(b)** \frac{7}{3} | Divide by $\cos\theta$ |
| 6 | **(b)** $90^\circ$ | Complementary |
| 7 | **(a)** $\frac{7}{6}$ | Substitute the standard values |
| 8 | **(c)** 5 | $\frac{c}{q}+\frac{p}{q}$ |
| 9 | **(a)** \frac{2}{3} | $\frac{c}{p}-\frac{q}{p}$ |
| 10 | **(d)** 2 | Each ratio is 1 |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Divide by $\cos A$: $\dfrac{\tan A+1}{\tan A-1}$ (1) $=7$ (1). | 1+1 |
| 12 | $\frac{1}{2}=\sin 30^\circ$ (1); $2x+0=30$ (1); $x=15^\circ$ (1). | 1+1+1 |
| 13 | Use $\sin^2 45^\circ=\frac{1}{2}$, $\cos^2 45^\circ=\frac{1}{2}$ (2); value $=1$ (1). | 2+1 |
| 14 | Divide by $\cos A$: $\dfrac{\tan A+1}{\tan A-1}$ (1) $=\frac{17}{7}$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $\frac{1}{2}=\sin 30^\circ$ (1); $3x+0=30$ (1); $x=10^\circ$ (1). | 1+1+1 |
| 16 | Hypotenuse $=\sqrt{15^2+8^2}=17$ (1); $\sin\theta=\frac{15}{17}$, $\cos\theta=\frac{8}{17}$ (1); $\sec\theta=\frac{17}{8}$ (1). | 1+1+1 |
| 17 | $\sqrt{\cos^2 A}=\cos A$ (1); $\cos A\cdot\sec A$ (1) $=1$ (1). | 1+1+1 |
| | **OR:** $\sqrt{3}=\tan 60^\circ$ (1); $2x+0=60$ (1); $x=30^\circ$ (1). | 1+1+1 |
| 18 | Hypotenuse $=\sqrt{24^2+7^2}=25$ (1); $\sin\theta=\frac{24}{25}$, $\cos\theta=\frac{7}{25}$ (1); $\sec\theta=\frac{25}{7}$ (1). | 1+1+1 |
| | **OR:** $\dfrac{\sin A}{\csc A}=\sin^2 A$, $\dfrac{\cos A}{\sec A}=\cos^2 A$ (2); sum $=1$ (1). | 2+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | LHS $=\left(\dfrac{1-\sin A}{\cos A}\right)^2$ (2); $=\dfrac{(1-\sin A)^2}{1-\sin^2 A}$ (1); $=\dfrac{1-\sin A}{1+\sin A}$ (2). | 5 |
| | **OR:** Divide numerator and denominator by $\cos A$ (1): $\dfrac{3\tan A+2}{4\tan A+1}$ (1); $=\dfrac{3(\frac{15}{8})+2}{4(\frac{15}{8})+1}=\frac{61}{68}$ (2); verified with $\sin A=\frac{15}{17}$, $\cos A=\frac{8}{17}$ (1). | 1+1+2+1 |
| 20 | LHS $=\dfrac{\cos^2 A+(1+\sin A)^2}{\cos A(1+\sin A)}$ (2); $=\dfrac{2+2\sin A}{\cos A(1+\sin A)}$ (2); $=\dfrac{2}{\cos A}=2\sec A$ (1). | 5 |
| | **OR:** Divide numerator and denominator by $\cos A$ (1): $\dfrac{2\tan A+5}{3\tan A+2}$ (1); $=\dfrac{2(\frac{24}{7})+5}{3(\frac{24}{7})+2}=\frac{83}{86}$ (2); verified with $\sin A=\frac{24}{25}$, $\cos A=\frac{7}{25}$ (1). | 1+1+2+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** \frac{17}{5} | Divide by $\cos\theta$ |
| 2 | **(d)** $90^\circ$ | Complementary |
| 3 | **(b)** $\frac{3}{2}$ | Substitute the standard values |
| 4 | **(b)** 4 | $\frac{c}{q}+\frac{p}{q}$ |
| 5 | **(b)** \frac{3}{5} | $\frac{c}{p}-\frac{q}{p}$ |
| 6 | **(d)** 2 | Each ratio is 1 |
| 7 | **(c)** \frac{23}{8} | Divide by $\cos\theta$ |
| 8 | **(b)** $90^\circ$ | Complementary |
| 9 | **(a)** $\frac{5}{4}$ | Substitute the standard values |
| 10 | **(b)** 7 | $\frac{c}{q}+\frac{p}{q}$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $\sqrt{3}=\tan 60^\circ$ (1); $3x+0=60$ (1); $x=20^\circ$ (1). | 1+1+1 |
| 12 | Use $\tan^2 60^\circ=3$, $\sin^2 30^\circ=\frac{1}{4}$ (2); value $=\frac{23}{4}$ (1). | 2+1 |
| 13 | Divide by $\cos A$: $\frac{\tan A+1}{\tan A-1}$ (1) $=\frac{31}{17}$ (1). | 1+1 |
| 14 | $\frac{1}{2}=\cos 60^\circ$ (1); $2x+10=60$ (1); $x=25^\circ$ (1). | 1+1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $\frac{\sqrt{3}}{2}=\cos 30^\circ$ (1); $3x+0=30$ (1); $x=10^\circ$ (1). | 1+1+1 |
| 16 | Hypotenuse $=\sqrt{21^2+20^2}=29$ (1); $\sin\theta=\frac{21}{29}$, $\cos\theta=\frac{20}{29}$ (1); $\sec\theta=\frac{29}{20}$ (1). | 1+1+1 |
| 17 | $1+\tan^2 A=\sec^2 A$ (1); $\cos^2 A\sec^2 A$ (1) $=1$ (1). | 1+1+1 |
| | **OR:** $\frac{1}{\sqrt{3}}=\tan 30^\circ$ (1); $2x+0=30$ (1); $x=15^\circ$ (1). | 1+1+1 |
| 18 | Hypotenuse $=\sqrt{40^2+9^2}=41$ (1); $\sin\theta=\frac{40}{41}$, $\cos\theta=\frac{9}{41}$ (1); $\sec\theta=\frac{41}{9}$ (1). | 1+1+1 |
| | **OR:** $\sin^2 A\cdot\frac{\cos^2 A}{\sin^2 A}=\cos^2 A$ (1); $\cos^2 A\tan^2 A=\sin^2 A$ (1); sum $=1$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | LHS $=\frac{\sec^2 A}{\csc^2 A}=\tan^2 A$ (2); RHS: $1-\cot A=\frac{\tan A-1}{\tan A}$ so $\frac{1-\tan A}{1-\cot A}=-\tan A$ (2); square $=\tan^2 A$ (1). | 5 |
| | **OR:** Divide numerator and denominator by $\cos A$ (1): $\frac{5\tan A+3}{1\tan A+1}$ (1); $=\frac{5(\frac{21}{20})+3}{1(\frac{21}{20})+1}=\frac{165}{41}$ (2); verified with $\sin A=\frac{21}{29}$, $\cos A=\frac{20}{29}$ (1). | 1+1+2+1 |
| 20 | LHS $=(\sin^2\theta-\cos^2\theta)(\sin^2\theta+\cos^2\theta)$ (2); $=\sin^2\theta-\cos^2\theta$ (1); $=\sin^2\theta-(1-\sin^2\theta)=2\sin^2\theta-1$ (2). | 5 |
| | **OR:** Divide numerator and denominator by $\cos A$ (1): $\frac{4\tan A+7}{2\tan A+1}$ (1); $=\frac{4(\frac{40}{9})+7}{2(\frac{40}{9})+1}=\frac{223}{89}$ (2); verified with $\sin A=\frac{40}{41}$, $\cos A=\frac{9}{41}$ (1). | 1+1+2+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Introduction to Trigonometry** | Class 10 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** \frac{3}{4} | $\frac{c}{p} - \frac{q}{p}$ |
| 2 | **(b)** 2 | Each ratio is 1 |
| 3 | **(d)** \frac{31}{7} | Divide by $\cos\theta$ |
| 4 | **(d)** $90^\circ$ | Complementary |
| 5 | **(b)** 0 | Substitute the standard values |
| 6 | **(c)** \frac{5}{2} | $\frac{c}{q} + \frac{p}{q}$ |
| 7 | **(d)** \frac{3}{7} | $\frac{c}{p} - \frac{q}{p}$ |
| 8 | **(b)** 2 | Each ratio is 1 |
| 9 | **(c)** \frac{41}{20} | Divide by $\cos\theta$ |
| 10 | **(d)** $90^\circ$ | Complementary |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Use $\cos^2 30^\circ = \frac{3}{4}$, $\sin^2 60^\circ = \frac{3}{4}$ (2); value $= \frac{3}{2}$ (1). | 2+1 |
| 12 | Divide by $\cos A$: $\frac{\tan A + 1}{\tan A - 1}$ (1) $= -\frac{17}{7}$ (1). | 1+1 |
| 13 | $1 = \tan 45^\circ$ (1); $5x + 0 = 45$ (1); $x = 9^\circ$ (1). | 1+1+1 |
| 14 | Use $\sin^2 60^\circ = \frac{3}{4}$, $\cos^2 30^\circ = \frac{3}{4}$ (2); value $= \frac{3}{4}$ (1). | 2+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $\frac{\sqrt{3}}{2} = \sin 60^\circ$ (1); $3x + 0 = 60$ (1); $x = 20^\circ$ (1). | 1+1+1 |
| 16 | Hypotenuse $= \sqrt{8^2 + 6^2} = 10$ (1); $\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$ (1); $\sec\theta = \frac{5}{3}$ (1). | 1+1+1 |
| 17 | Use $(a-b)(a+b) = a^2 - b^2$ (1); $\csc^2 A - \cot^2 A$ (1) $= 1$ (1). | 1+1+1 |
| | **OR:** $\frac{1}{\sqrt{2}} = \cos 45^\circ$ (1); $5x + 0 = 45$ (1); $x = 9^\circ$ (1). | 1+1+1 |
| 18 | Hypotenuse $= \sqrt{12^2 + 9^2} = 15$ (1); $\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$ (1); $\sec\theta = \frac{5}{3}$ (1). | 1+1+1 |
| | **OR:** $\frac{1}{\sec^2 A} + \frac{1}{\csc^2 A}$ (1); $= \cos^2 A + \sin^2 A$ (1) $= 1$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | LHS $= \frac{1 + \frac{1}{\cos A}}{\frac{1}{\cos A}} = \cos A + 1$ (2); RHS $= \frac{1 - \cos^2 A}{1 - \cos A}$ (2); $= 1 + \cos A$ (1). | 5 |
| | **OR:** Divide numerator and denominator by $\cos A$ (1): $\frac{6\tan A + 1}{3\tan A + 1}$ (1); $= \frac{6(\frac{4}{3}) + 1}{3(\frac{4}{3}) + 1} = \frac{9}{5}$ (2); verified with $\sin A = \frac{4}{5}$, $\cos A = \frac{3}{5}$ (1). | 1+1+2+1 |
| 20 | LHS $= \sqrt{\frac{(1+\sin\theta)^2}{1 - \sin^2\theta}}$ (2); $= \frac{1 + \sin\theta}{\cos\theta}$ (2); $= \sec\theta + \tan\theta$ (1). | 5 |
| | **OR:** Divide numerator and denominator by $\cos A$ (1): $\frac{1\tan A + 8}{2\tan A + 1}$ (1); $= \frac{1(\frac{4}{3}) + 8}{2(\frac{4}{3}) + 1} = \frac{28}{11}$ (2); verified with $\sin A = \frac{4}{5}$, $\cos A = \frac{3}{5}$ (1). | 1+1+2+1 |

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